

Christos Iliakopoulos

Curriculum Vitae

National Technical University of Athens
Fifth Year Undergraduate Student
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Research Interests

- Computer Networks
- Impact of AI and ML on Computer Networks
- Computer Science and Software Engineering Technologies

Education

- Present **National Technical University of Athens**
Major: Electrical and Computer Engineering
- 2016 – 2019 **The Moraitis Private School, Athens, Greece**

Relevant Coursework

- **Computing Systems:** Introduction to Computer Programming (3020), Programming Techniques and Data Structures (3138), Fundamentals of Computer Science (3355), Computer Architecture (3357), Computer Operating Systems (3136), Human-Computer Interaction (3362), Software Engineering (3205), Multimedia Technology (3297)
- **Networks:** Queueing Systems (3248), Digital Communications I (3299), Computer Networks (3298), Computer Networks Laboratory (3278), Optical Communication Networks (3367)
- **ML/AI:** Machine Learning (3403), Artificial Intelligence (3287)
- **Mathematics and Physics:** Calculus I & II (3131 - 3137), Linear Algebra (3132), Differential Equations (3374), Physics I (3049)

Technical Skills

Programming languages: Python, C/C++, Java, JavaScript, Dart, Mips Assembly, Matlab

Programming frameworks: Scikit-learn, Flutter

Back-end Technologies: Node.js, MySQL, Express.js

Networks: Routing Algorithms and Protocols (RIP, OSPF, BGP); Flow Control Algorithms (TCP Tahoe, Reno, Leaky Bucket Algorithm); Flow and Traffic Scheduling Algorithms; Firewalls and NAT; Queueing Theory Algorithms;

Projects

- **Computer Networks Laboratory.** During my coursework in Computer Networks, I gained extensive hands-on experience configuring and troubleshooting network systems using VirtualBox and FreeBSD. I worked with key networking concepts, including LAN setup, ARP, bridges, and both static and dynamic routing protocols such as RIP, OSPF, and BGP, across IPv4 and IPv6 environments. Using tools like tcpdump and ipfw, I analyzed network traffic, implemented firewalls, and configured NAT to secure and manage traffic flow. I also learned to use Quagga and FRRouting (FRR) for advanced

routing. I configured DHCP and DNS for assigning IPs and resolving names, worked with IPv6 features like Neighbor Discovery Protocol, and set up and utilized basic internet services (DHCP, DNS, HTTP server, NAT) .

- **Software Engineering Project - Toll Interoperability.** As part of our Software Engineering Project: Toll Interoperability, my colleagues and I developed a robust, secure, and scalable system to facilitate seamless toll station operations. The project included the creation of a back-end system using Node.js, Express, and TypeScript, coupled with a PostgreSQL database to manage and store essential data such as toll passes, operators, stations, and user accounts. The front-end, built with HTML, CSS, JavaScript, and EJS templates, provided a user-friendly interface for accessing various features. To ensure secure communication, we implemented SSL certificates, and authentication mechanisms using custom headers like X-OBSERVATORY-AUTH for user sessions.

Additionally, we incorporated an administrator panel to allow management of system users, passes, and stations through both a web interface and a CLI client. A comprehensive testing suite, utilizing shell scripts and automated testing frameworks, validated the functionality of all API endpoints and ensured the system's reliability. By leveraging modern web technologies and a well-structured development approach, we successfully delivered a solution that meets the interoperability needs of toll operators while providing robust analytics, reporting, and seamless data handling.

- **Human and Computer Interaction - SensorTunes.** For our HCI project, my colleagues and I developed a modern mobile application designed to showcase innovation through advanced design and sensor integration. The project, implemented in Flutter and prototyped in Figma, focused on creating an intelligent MP3 player application. The app leverages smartphone sensors to adapt to the user's environment dynamically. For instance, using the front camera, it adjusts its display when the ambient lighting changes, ensuring optimal visibility. Similarly, when traveling by car, the application automatically activates a "Driving Mode," streamlining controls to minimize distractions and enhance safety.

This project allowed us to explore user-centered design principles by delivering an alpha prototype in Figma and evolving it into a functional app in Flutter. It combined the disciplines of affective computing, user adaptation, and mobile interactivity to offer an engaging user experience. Through collaborative teamwork, we successfully integrated technical capabilities with a polished design, meeting the project's objectives of usability and innovation.

- **Operating Systems.** In my Operating Systems course, I worked independently on four assignments that delved into key concepts such as process management, signal handling, inter-process communication, and client-server communication using sockets in Linux. Here's a summary of each assignment:
 - **Assignment 1:** Developed a C program to demonstrate process creation using `fork()`, file manipulation with `open` and `write` functions, and process identification with `getpid` and `getppid`. The task involved creating a parent and a child process, where both wrote their process details (PID and PPID) to the same specified file. The child process completed its task and terminated first, after which the parent process proceeded to write its data and terminate in a controlled manner.
 - **Assignment 2:** Developed a program to simulate a hierarchical tree of processes representing "gates" in a multi-level structure. Each gate process could be dynamically controlled through signals such as `SIGUSR1`, `SIGUSR2`, and `SIGTERM`, which were handled using `sigaction()` for custom signal processing. `SIGUSR1` and `SIGUSR2` were used to toggle or activate specific gate behaviors, while `SIGTERM` was employed to gracefully terminate processes. This hierarchical model

showcased parent-child relationships and inter-process communication in Linux. A Makefile was utilized to streamline the compilation and execution process, ensuring modularity and ease of use.

- **Assignment 3:** This assignment focused on inter-process communication and synchronization using pipes in Linux. The task was to implement a program where a parent process creates multiple child processes (n children) and communicates with them through pipes. The parent process could receive commands from the terminal, such as assigning tasks (sending integers to a specific child process) and terminating all child processes. Using the `select()` system call, the parent process monitored multiple file descriptors to manage input from both the terminal and the child processes. Each child process, upon receiving a task from the parent through a pipe, incremented the received value, simulated workload with a delay, and sent the result back to the parent via another pipe. The program supported two task assignment methods: *round-robin* and *random*. It also included robust error handling and dynamic memory allocation for managing processes and pipes.
- **Assignment 4:** Developed a client-server communication system using sockets in C. The task involved building a client program capable of sending commands to a server, retrieving data, and parsing server responses. The client implemented a user-friendly menu for interacting with the server, supporting commands like querying specific information, adding entries, or retrieving aggregated results. The communication was implemented using TCP sockets to ensure reliable data transfer. The client dynamically processed server responses, displayed results clearly, and handled errors like connection failures, invalid commands, and unexpected server issues. The project demonstrated practical implementation of networking concepts, emphasizing communication protocols like TCP, data parsing, and user interaction within a client-server architecture.
- **Multimedia Technology - Library Management System.** In my Multimedia Technology course, I developed a Library Management Application using JavaFX and Java SDK. The application allowed for seamless book management, borrowing, and user interaction. Administrators could add, modify, and remove books, users and categories, with changes reflecting instantly through implemented listeners. Users could search for books, borrow up to two books for five days, and rate or comment on them. The project employed object serialization for data persistence, storing user, book, category, and borrowing data in `.ser` files. All functionalities were controlled by dedicated managers and FXML controllers, ensuring a robust implementation of object-oriented principles. Additionally, a graphical user interface was created to enhance usability, leveraging SceneBuilder and CSS for styling. The application incorporated debugging logs for confirmation of object states and included a pre-configured admin user for ease of access during demonstrations.

Competitions

I participated in the IEEE Xtreme 18.0 programming competition as a member of a team. Throughout the 24-hour contest, my team and I consistently ranked among the top 100 teams worldwide for the majority of the competition.

Volunteering Activities

I was part of the electrical engineering team within the Unique Minds organization, where we implemented a project utilizing an LM3914 LED bar graph. The LED bar was controlled by a potentiometer, allowing students to adjust the brightness of the LEDs by regulating the voltage drawn from a battery. Our objective was to guide younger students through the principles of circuit design and operation, demonstrating how components like integrated circuit (LM3914), potentiometer and LEDs interact in a practical application while fostering their interest in electrical engineering.

Miscellaneous

I am fluent in Greek and English and enjoy exploring creative hobbies like video and photo editing. I have a strong passion for volleyball, having competed semi-professionally with notable achievements such as 1st place in the Panhellenic School Championship and 2nd place in the Under-21 category. I also love traveling and exploring the world. In my free time, I enjoy reading books, particularly crime novels.